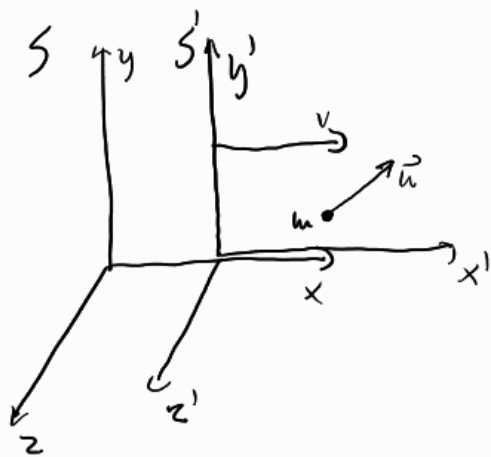




odsllej $m_0 \rightarrow m$

v S:

$$E = \gamma_u m c^2$$

$$p_x = \gamma_u m u_x$$

$$p_y = \gamma_u m u_y$$

$$p_z = \gamma_u m u_z$$

v S':

$$E' = \gamma_{u'} m c^2$$

$$p_x' = \gamma_{u'} m u_x'$$

$$p_y' = \gamma_{u'} m u_y'$$

$$p_z' = \gamma_{u'} m u_z'$$

$$\gamma_{u'} = \left(1 - \frac{u_x'^2 + u_y'^2 + u_z'^2}{c^2}\right)^{-1/2}$$

$$= \left(1 - \frac{(u_x - v)^2}{\left(1 - \frac{u_x v}{c^2}\right)^2 c^2} - \frac{u_y^2 \left(1 - \frac{v^2}{c^2}\right)}{\left(1 - \frac{u_x v}{c^2}\right)^2 c^2} - \frac{u_z^2 \left(1 - \frac{v^2}{c^2}\right)}{\left(1 - \frac{u_x v}{c^2}\right)^2 c^2}\right)^{-1/2}$$

$$= \left(\frac{c^2 \left(1 - \frac{u_x v}{c^2}\right)^2 - (u_x - v)^2 - u_y^2 \left(1 - \frac{v^2}{c^2}\right) - u_z^2 \left(1 - \frac{v^2}{c^2}\right)}{c^2 \left(1 - \frac{u_x v}{c^2}\right)^2}\right)^{-1/2}$$

$$= \left(\frac{c^2 - 2u_x v + \overbrace{\frac{u_x^2 v^2}{c^2}}^{= u_x^2 \left(1 - \frac{v^2}{c^2}\right)} - u_x^2 + 2u_x v - v^2 - u_y^2 \left(1 - \frac{v^2}{c^2}\right) - u_z^2 \left(1 - \frac{v^2}{c^2}\right)}{c^2 \left(1 - \frac{u_x v}{c^2}\right)^2}\right)^{-1/2}$$

$$= \left(\frac{c^2 - v^2 - u^2 \left(1 - \frac{v^2}{c^2}\right)}{c^2 \left(1 - \frac{u_x v}{c^2}\right)^2}\right)^{-1/2} = \gamma_v \frac{1 - \frac{u_x v}{c^2}}{\sqrt{1 - \frac{u^2}{c^2}}} = \boxed{\gamma_v \gamma_u \left(1 - \frac{u_x v}{c^2}\right) = \gamma_{u'}}$$

$$E' = \gamma_{u'} m c^2 = \gamma_v \left[\underbrace{\frac{m c^2}{\sqrt{1 - \frac{u^2}{c^2}}}}_E - \underbrace{\frac{m u_x v}{\sqrt{1 - \frac{u^2}{c^2}}}}_{p_x} \right]$$

$$\boxed{E' = \gamma_v (E - v p_x)}$$

$$p'_x = \gamma_{u'} m u'_x = \gamma_v \gamma_u \left(1 - \frac{u_x v}{c^2}\right) m \frac{u_x - v}{1 - \frac{u_x v}{c^2}} = \gamma_v \left[\underbrace{\frac{m u_x}{\sqrt{1 - \frac{u^2}{c^2}}}}_{p_x} - \underbrace{\frac{m v}{\sqrt{1 - \frac{u^2}{c^2}}}}_{\frac{E}{c^2}} \right] = \gamma_v \left(p_x - \frac{E v}{c^2} \right)$$

$$\boxed{\begin{aligned} p'_x &= \gamma_v \left(p_x - \frac{E v}{c^2} \right) \\ p'_y &= p_y \\ p'_z &= p_z \end{aligned}}$$

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